

Sadhana Jamkatel

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EDUCATION

North Central College, Naperville, IL

Expected: May 2027

Major: Bachelor of Science in Computer Science, **Minor:** Economics

GPA 3.87/4

Relevant Coursework: Data Structures, Computer Organization/Design, Software Development, Database System, Algorithm & Design, Computer Network and Security, Algorithms and Design

WORK EXPERIENCE

DevOps Engineer Intern

Jun 2025-Aug 2025

Aeropay

- Reduced documentation search time from 30 minutes to 30 seconds by building a full-stack internal documentation site using Docusaurus (MDX, custom charts, integrated search), serving 30+ engineers daily and hosted on GitHub Pages with automated CI/CD via GitHub Actions ; also tested the site using AWS S3 + CloudFront with JumpCloud SSO access controls
- Created a documentation site for a CLI tool (used for automating login, role-based access, and token generation for AWS and databases) that auto-generates and updates files using OCLIF's markdown generator and GitHub Actions.

High-Performance Computing (HPC) Bootcamp Participant

Aug 2025

Argonne National Laboratory

- Developed HPC chatbot for scientific Q&A using LangChain, HuggingFace, and Gemini API with RAG architecture, achieving 35% accuracy improvement and 45% latency reduction (3.2s to 1.8s) compared to non-RAG baseline

PROJECTS

NVIDIA Jetson AGX Thor Object Detection (Python, OpenCV, YOLOv8)

Jan 2026- Present

- Implemented an embedded computer vision system on NVIDIA Jetson AGX Thor using Python, OpenCV, and YOLOv8 to process real-time camera input, preparing the system for automated dataset analysis
- Configured headless Linux environment, virtual environments, and project workflow for efficient development, testing, and reproducible results.
- Validated object detection on live and pre-recorded inputs, and automated data transfer via SCP, laying groundwork for future model training and performance evaluation.

Care Equity | Full-Stack Healthcare Bias Tracker

Feb 2026

Hackathon Code2040

- Engineered a full-stack Next.js and Node.js application to surface maternal healthcare inequities, featuring a responsive hospital discovery tool, interactive data visualizations (Recharts), and an anonymous reporting system.
- Designed a robust REST API with Express and MongoDB, implementing a custom same-origin API proxy to resolve cross-browser and mobile compatibility issues, ensuring seamless data delivery across all devices.
- Managed end-to-end deployment using Vercel (Frontend) and Render (Backend), configuring environment-based routing and automated "keep-awake" strategies to maintain high availability for production users.

Predictive Maintenance ML System for Fleet Engine Monitoring

Sep 2025

- Reduced engine downtime by 95% by building a predictive maintenance system using Random Forest to analyze 21 sensor readings from NASA Turbofan data and detect at-risk engines before failure.
- Designed React/Typescript dashboard with alerts and risk visualization to monitor 100 engines and notify maintenance teams of critical issues.
- Achieved 96% accuracy with 4.28% false alarm rate, delivering \$750K+ projected annual cost savings through optimized hyperparameter tuning and feature engineering.

ON CAMPUS INVOLVEMENT

Resident Assistant | North Central College

May 2024 – Present

- Increased resident engagement by 95% through organizing 5+ community-building and professional development events per semester
- Provided mentorship and academic support to **40+ residents**, fostering an inclusive and supportive living environment

Vice President | Society of Hispanic Professional Engineers (SHPE)

Aug 2025 – Present

- Obtained college funding enabling 6+ members to attend national engineering conferences
- Strengthened industry engagement by establishing connections with external companies, creating opportunities for students to explore professional engineering environments and directed careers workshops and streamlined executive board coordination to improve organizational efficiency

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C/C++, Java

Web & Frameworks: React.js, Node.js, Flask, Next.js, HTML/CSS

Cloud & DevOps: AWS S3, EC2, CloudFront, CloudWatch, Docker, GitHub Actions, Linux/Unix, Kubernetes, Salesforce Sales Cloud, Salesforce CRM

Data & ML: PostgreSQL, MongoDB, scikit-learn, pandas, NumPy, LangChain, SQL, GenAI tools, Yolo, OpenCV